

**Problem 1:** Take 4 counters and put them all in one pile. The move: take one counter from every pile, and put all the counters you took together as one new pile. A pile with no counters left is gone. After each move, put the piles in order from biggest to smallest and write the pile sizes in the next box, like 3 1. Keep going until you write an arrangement that is already in the strip, and circle the two matching boxes. Then do the same starting from 2 2 and from 1 1 1 1. In the last row, write the loop that all three runs fall into, in order, and draw an arrow from the last box back to the first. In the circle, write how many arrangements are in the loop.

|         |   |  |   |  |   |  |   |  |   |   |
|---------|---|--|---|--|---|--|---|--|---|---|
| 4       | → |  | → |  | → |  | → |  | → |   |
| 2 2     | → |  | → |  | → |  | → |  |   |   |
| 1 1 1 1 | → |  | → |  | → |  | → |  | → |   |
|         | → |  | → |  | → |  | → |  |   | ○ |

**Problem 2:** Take 6 counters, all in one pile, and run the move. This time keep going until a move gives back exactly the arrangement you just wrote, and then stop. An arrangement that the move does not change is called a stop. Circle the stop.

|   |   |  |   |  |   |  |   |  |   |  |
|---|---|--|---|--|---|--|---|--|---|--|
| 6 | → |  | → |  | → |  | → |  | → |  |
|---|---|--|---|--|---|--|---|--|---|--|

**Problem 3:** Find all the different ways to put 6 counters in piles, and write each one in its own box, biggest pile first. Two ways with the same sizes in a different order count as the same way. Write how many ways you found in the circle. Then write on the lines how you organized your search, so that you can be sure no way is missing.

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |




---



---

**Problem 4:** Run the move from every way in Problem 3, one strip for each way, and share the strips among your table. In each strip, circle the stop the first time it appears, and write the number of moves before it in the small circle at the end.

**Problem 5:** Make a map. Write each way from Problem 3 in the column for its number of moves from Problem 4, spread out so that there is room around it. Then draw an arrow from each arrangement to the arrangement it becomes after one move. (a) Which arrangements have no arrow pointing to them? (b) How many arrows point to 4 2, to 5 1, and to 3 2 1? (c) Follow the arrows from the arrangement farthest from the stop. Write that longest chain.

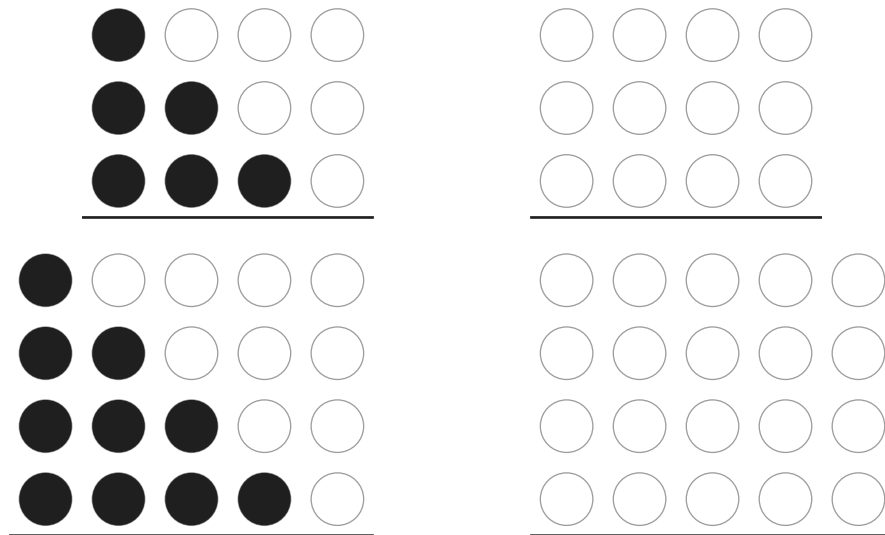
| 6 moves | 5 moves | 4 moves | 3 moves | 2 moves | 1 move | 0 moves |
|---------|---------|---------|---------|---------|--------|---------|
|         |         |         |         |         |        |         |

---

---

---

**Problem 6:** Here is 3 2 1 drawn as towers of dots, one tower for each pile, tallest first. The move takes the top dot off every tower, and those dots become the new pile. Cross out the top dot of each tower and draw the new pile as a tower of dots in the empty column. In the second picture, draw the result with its towers in order, tallest first. Explain to your table why the move can never change 3 2 1. Do the same for 4 3 2 1 in the third and fourth pictures. Then build 5 4 3 2 1 with 15 counters and check it the same way.



**Problem 7:** Suppose some arrangement with exactly 3 piles is a stop. (a) How many counters does the move put in the new pile? (b) After the move, which pile is the biggest, and why? (c) So how big is the biggest pile of the stop? (d) After the move, which pile is the second biggest? So how big is the second biggest pile of the stop? (e) Write the only arrangement with 3 piles that can be a stop. (f) Do the same for 4 piles and for 5 piles, and write the only possible stops. (g) Circle every number from 1 to 30 that is the total number of counters in some stop.

---



---



---



---



---

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |

**Problem 8:** Go backwards. (a) Find an arrangement of 6 counters that becomes 3 3 after one move. (b) Find every arrangement that becomes 4 2 after one move. (c) Find an arrangement that becomes 2 2 1 1 after one move, or explain to your table why there is none. (d) Use your map from Problem 5: for each of 6, 5 1, 2 2 2 and 3 2 1, write how many arrangements become it after one move. (e) For each of the three arrangements with no arrow pointing to them, write its biggest pile and its number of piles. Do the same for three arrangements that do have an arrow pointing to them. Compare the two lists.

---



---



---



---



---



---

**Problem 9:** Here are all 15 ways to put 7 counters in piles. Predict which of them nothing becomes after one move, and circle your predictions. Then check every one of the 15 with counters: for each way, find an arrangement that becomes it, or convince your table that none exists.

|           |         |           |             |               |
|-----------|---------|-----------|-------------|---------------|
| 7         | 6 1     | 5 2       | 5 1 1       | 4 3           |
| 4 2 1     | 4 1 1 1 | 3 3 1     | 3 2 2       | 3 2 1 1       |
| 3 1 1 1 1 | 2 2 2 1 | 2 2 1 1 1 | 2 1 1 1 1 1 | 1 1 1 1 1 1 1 |

---



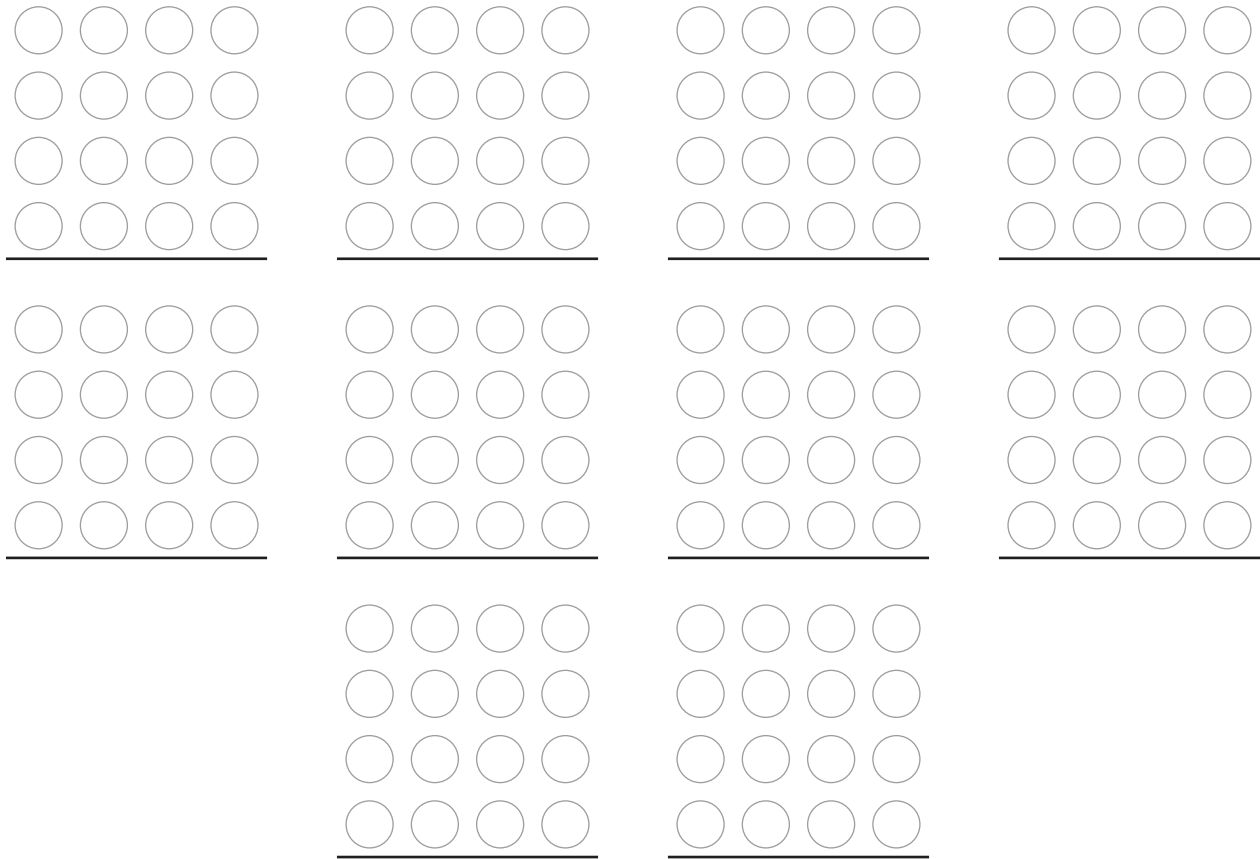
---

**Problem 10:** Loops. Start from 5 and run the move until an arrangement repeats, then start from 2 2 1. Write the loop for 5 counters in the loop row. Then do the same for 7 counters starting from 7, and for 8 counters starting from 8 and then from 5 3. Write each loop in a loop row, with the number of arrangements in its circle.

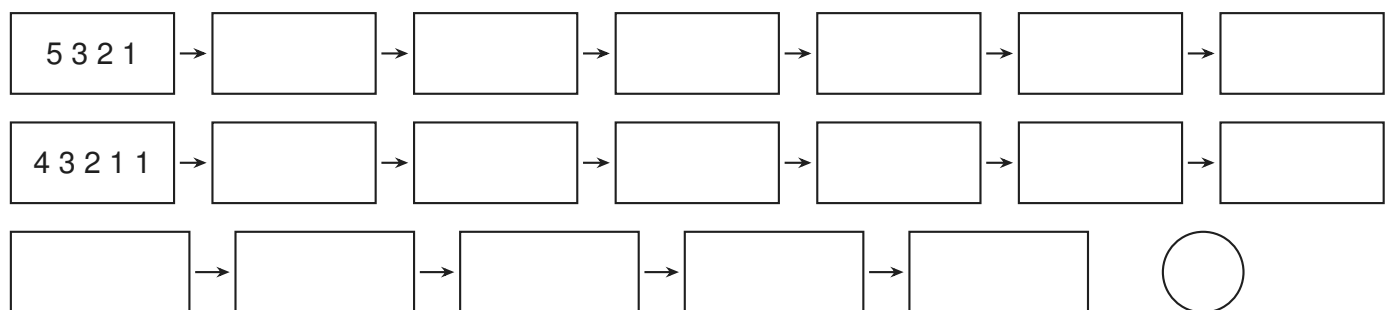
|       |   |  |   |  |   |  |   |  |   |  |   |  |
|-------|---|--|---|--|---|--|---|--|---|--|---|--|
| 5     | → |  | → |  | → |  | → |  | → |  | → |  |
| 2 2 1 | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |
| 7     | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |
| 8     | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |
| 5 3   | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |
|       | → |  | → |  | → |  | → |  | → |  | → |  |



**Problem 11:** Draw each arrangement of the loop for 7 counters as towers of dots, in the order they come. In each picture, color the stairs 3 2 1 in one color and the one counter left over in another color. Say where the leftover counter goes from one picture to the next. Then draw the four arrangements of the long loop for 8 counters and the two arrangements of the short loop, and color the stairs 3 2 1 and the two leftover counters the same way.



**Problem 12:** Take 11 counters: the stairs 4 3 2 1 and one extra counter. Start from 5 3 2 1 and run the move until an arrangement repeats. Then start from 4 3 2 1 1. Write the loop for 11 counters and how many arrangements are in it.



**Problem 13:** Take 10 counters. Run the move from 10 in the first strip and write the number of moves to the stop in its circle. Then find a start that takes more moves than that, and then one that takes more than 9 moves. Write your best runs in the other strips, with the number of moves in the circles.

**Problem 14:** With 3 counters there are only three starts; find the most moves any of them needs to reach the stop 2 1. Write that number, the most for 6 counters from Problem 4, and the most you found for 10 counters, on the line. Guess the most for 15 counters. Then start from 4 4 3 2 1 1 with 15 counters, run the move to the stop, and write the number of moves in the circle.

A sequence of boxes connected by arrows, starting with the number 443211, followed by seven empty boxes, and ending with a circle.